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BCvoteMDE: A Blockchain-l Scheme for Multi-Distric

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***Abstract*—The traditional electronic voting (e-voting) has the problems of dead vote, repetition and missing registration, which cannot reflect the voting result correctly, objectively and fairly. With the booming development of blockchain in recent years, blockchain technology provides a new solution in the field of e-voting. Multi-District election is an important method of election in real life, e.g. in US presidential election. Inspired by the electoral college system, we propose a blockchain-based e-voting scheme for multi-district election. First, we propose a blockchain-based voter registration method that all voters are authenticated by the blockchain system. Second, we design a two-layer blockchain architecture, where the lower layer records the votes of voters in each district and the higher layer records the votes of electors. Then we describe the voting process and evaluate the security and availability of the proposed scheme. The experimental results show that the proposed scheme can satisfy the needs of multi-district election.**

***Keywords*—Blockchain, Electronic Voting, Multi-District Elections.**

I. Introduction

Election is a form of procedure with recognized rules which is almost conducted through voting. Given the importance of elections to society, efforts have been made to optimize the reliability and efficiency of elections. With the development of technology, voting has evolved from traditional paper-based voting to electronic voting. Traditional Paper-based voting system used paper ballots for voting and counted by hand, while the E-voting system used the electronic voting in an Internet environment. The efficiency of voting has been increasing while the cost of voting has been decreasing. However, both paper-based and electronic voting are centralized voting systems, the centralized structure suffers from single point of failure as well as can lead to trust issues, such as double voting, dead ballots, lost ballots, deliberate delays in counting, inefficiency tallying, etc. In addition, the ballots recorded in traditional voting systems can be easily changed or even deleted. A real example is the South Wales (NSW) Electoral Commission's admission that the online iVote website [1] leaked as many as 66,000 electoral votes.

Blockchain technology brings a new solution to electronic voting systems. Blockchain has received considerable attention since its introduction in 2008 [2]. Blockchain is a distributed ledger, which is decentralized, tamper-proof and traceable. Initially used in the field of cryptocurrency, blockchain has since been applied to a variety of fields such as supply chain [3], healthcare [4], IOT [5], and more. Exploiting these exclusive properties, blockchain are also suitable for application to the field of electronic voting. It is believed that applying blockchain to the field of e-voting has great prospects. Electronic voting schemes based on

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